LavaFût TL-6 Drum Wash Tunnel – Site Acceptance Test Report

Coopérative acéricole Haute-Chaudière · Saint-Georges depot, Building C · CR-TL6-2026-04 rev. B · Issued 18 September 2026

1. PURPOSE AND SUMMARY

This report records the site acceptance test (SAT) of the LavaFût TL-6 drum wash tunnel installed in Building C of the cooperative's depot at 4870 rue des Érables, Saint-Georges (Beauce). The tunnel cleans and steam-treats the 205-litre steel drums that member producers return after delivering bulk maple syrup. It replaces the manual wash station in bay C-2, which was used for the last time on 29 August 2026 and is to be dismantled in October. The tunnel was supplied by Équipements Thibodeau-Rioux, which ran the factory acceptance test (FAT) at its Drummondville plant on 14 and 15 July 2026.

SAT runs took place from 8 to 12 September 2026 with 412 returned drums from 37 member producers. **Result: accepted with reservations.** Ten of the eleven acceptance tests passed; test SAT-09 (water consumption) failed and is tracked as punch item P-02. The tunnel is released for production from 22 September 2026, subject to the conditions in section 5. Revision B corrects the ATP swab values for drum batch 6 that were transposed in revision A of 16 September.

2. INSTALLED CONFIGURATION

- Tunnel serial number `TL6-0417`, five zones, chain conveyor with drums carried bung-down.
- PLC program `TL6-PLC v3.2.1`, loaded on 5 September 2026 (the FAT was run with `v3.1.4`; v3.2.1 adds the steam-zone door interlock and the French/English operator screens).
- Wash chemical: `ALC-15` alkaline detergent dosed to 1.5 % by a peristaltic pump; rinse water from the depot softener, hardness below 60 mg/L $CaCO_3$.
- Steam from the existing 300 kW boiler in room C-104 through a new reducing station set to 200 kPa.

TABLE 1. ZONE SET POINTS AND MEASURED VALUES DURING SAT (MEAN OF 412 DRUMS)

Zone	Function	Set point	Acceptance band	Measured mean	Result
Z1	Cold pre-rinse (syrup removal)	18 °C, 450 kPa	15–25 °C	19.4 °C	Pass
Z2	Alkaline wash, ALC-15 at 1.5 %	72 °C	68–76 °C	71.2 °C	Pass
Z2	Wash spray pressure	1 800 kPa	1 600–2 000 kPa	1 735 kPa	Pass
Z3	Hot rinse	85 °C	82–88 °C	83.6 °C	Pass
Z4	Steam treatment, interior	105 °C, 40 s	≥ 100 °C for ≥ 35 s	103.8 °C, 41 s	Pass
Z5	Hot-air drying	90 °C	85–95 °C	88.9 °C	Pass
–	Conveyor index time	52 s	48–56 s	53 s	Pass

3. ACCEPTANCE TESTS

Criteria come from the purchase specification `SPEC-ACH-2025-11` of 3 November 2025. Hygiene tests used ATP swabs taken inside the drum at the bung opening and the seam; residual sugar was measured with a refractometer on 250 mL of final rinse water shaken in the drum.

Note. Drums with visible crystallised syrup or a dented bung flange are not part of the SAT sample. Operators set them aside on the red rack for hand cleaning or scrapping; 23 such drums were removed during the test week.

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TABLE 2. SAT RESULTS AGAINST ACCEPTANCE CRITERIA

Test	Description	Criterion	Result	Status
SAT-01	Throughput, 4-hour run	≥ 60 drums/h	66 drums/h	Pass
SAT-02	Residual sugar in final rinse	≤ 0.2 °Brix	0.1 °Brix	Pass
SAT-03	ATP swab, bung opening	≤ 150 RLU	62 RLU	Pass
SAT-04	ATP swab, bottom seam	≤ 150 RLU	118 RLU	Pass
SAT-05	pH of final rinse (no detergent carry-over)	6.5–8.5	7.6	Pass
SAT-06	Residual water after drying	≤ 20 mL/drum	12 mL/drum	Pass
SAT-07	Emergency stop, conveyor halt	≤ 1.0 s	0.6 s	Pass
SAT-08	Steam-zone door interlock	No steam with door open	10/10 trials	Pass
SAT-09	Water consumption, Z1 + Z3	≤ 38 L/drum	43.5 L/drum	Fail
SAT-10	Noise at operator station	≤ 85 dBA	81 dBA	Pass
SAT-11	Detergent use, ALC-15	≤ 0.30 L/drum	0.24 L/drum	Pass

4. OPEN ITEMS

Items are classed A (blocks production), B (must be closed before the 2027 sugaring season) or C (improvement). No class A item remains open: item P-01, a leaking Z2 pump seal found on 8 September, was replaced by the supplier on 9 September and closed after the repeat run.

TABLE 3. PUNCH LIST AT ISSUE OF THIS REPORT

Item	Description	Class	Owner	Due date	Status
P-01	Z2 pump seal leak	A	Équipements Thibodeau-Rioux	9 Sep 2026	Closed
P-02	Water use 43.5 L/drum (SAT-09); fit flow restrictors and re-time Z1 nozzles	B	Équipements Thibodeau-Rioux	30 Nov 2026	Open
P-03	Guard at conveyor outfeed 40 mm too short	B	Gaétan Poulin, depot maintenance	16 Oct 2026	Open
P-04	Wash procedure and lock-out sheet to be issued	B	Raphaël Turcotte, quality	2 Oct 2026	Open
P-05	Condensate drain from Z4 routed to floor gully	B	Josée Lachance, site services	23 Oct 2026	Open

5. CONDITIONS OF RELEASE

1. Until P-02 is closed, readings of the rinse water flow meter `FT-101` are logged at the end of every shift, and a retest of SAT-09 on at least 100 drums is held within 10 working days of the modification.
2. Only operators trained by the supplier on 17 September 2026 (six persons, list kept by the quality office) may run the tunnel until the procedure of item P-04 is issued.
3. Washed drums are released to members only after a daily ATP swab on one drum per 50 washed reads ≤ 150 RLU; a reading above 300 RLU stops the line until the quality lead approves restart.
4. The final acceptance certificate and the release of the 10 % retention (CAD 48 600 of the CAD 486 000 contract) follow closure of all class B items.

Sign-off. Test lead: Mélanie Gosselin-Paré, ing., commissioning engineer, Coopérative acéricole Haute-Chaudière. Supplier: Félix Bergeron, site manager, Équipements Thibodeau-Rioux. Approval for production release: Raphaël Turcotte, quality lead, on 18 September 2026.

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